Database
(SOC-3020)

Alessandro Agostini[1,2]

aagostini@inha.ac.kr
agostini@disi.unitn.it

[1]LDKR-Group, School of Computer and Information Engineering
Inha University in Tashkent (IUT), Uzbekistan

[2]DISI, University of Trento, Italy

WEEK 3





DATABASE

Database Design: The E-R Data Model

WEEK 3

Code: W3.L1L2[LAB#1]

Outline

1. From Previous Week
2. Conceptual Data Design
3. Data and Data Relationships
4. Data Semantics
5. Data Constraints
6. Entity Integrity *Revised*: Weak Entities
7. Example: Company DB
8. LAB #1 : E-R Diagramming



Any question from you?

...

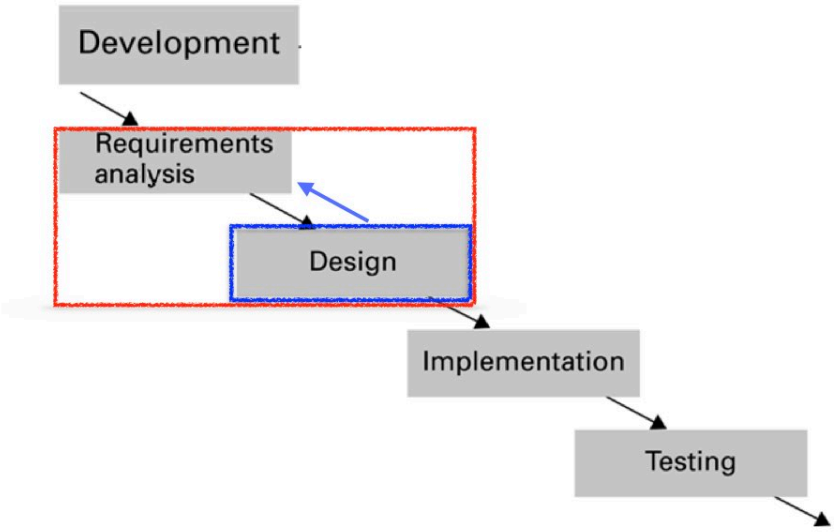
Outline

1. From Previous Week
2. Conceptual Data Design
3. Data and Data Relationships
4. Data Semantics
5. Data Constraints
6. Entity Integrity *Revised*: Weak Entities
7. Example: Company DB
8. LAB #1 : E-R Diagramming

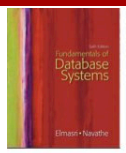
Conceptual Data Design

Phase 2 of Database System Development

Designer in Software Life Cycle







DB Design: Main Steps

- Heart of the database development process:

1. **Requirements collection and analysis (Phase 1)**
2. **Conceptual database design (Phase 2)**
3. **Choice of DBMS**
4. **Logical design (or: data model mapping) (Phase 4)**
5. **Physical design (Phase 5)**
6. **System implementation and tuning (“re-design”)**



List of Sentences

- After the Requirements Phase you shall have a list of sentences of this form:

  NOUN (“SUBJECT”) – VERB – NOUN (“OBJECT”)

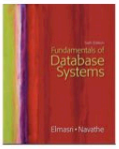
  about the application scenario on target, where a NOUN represents a set of “entities” that you need information about (**by querying**), and a VERB (or verb phrase, for example: *member of*, *for*, *seat-on*, etc.) represents “relationships” between entities.

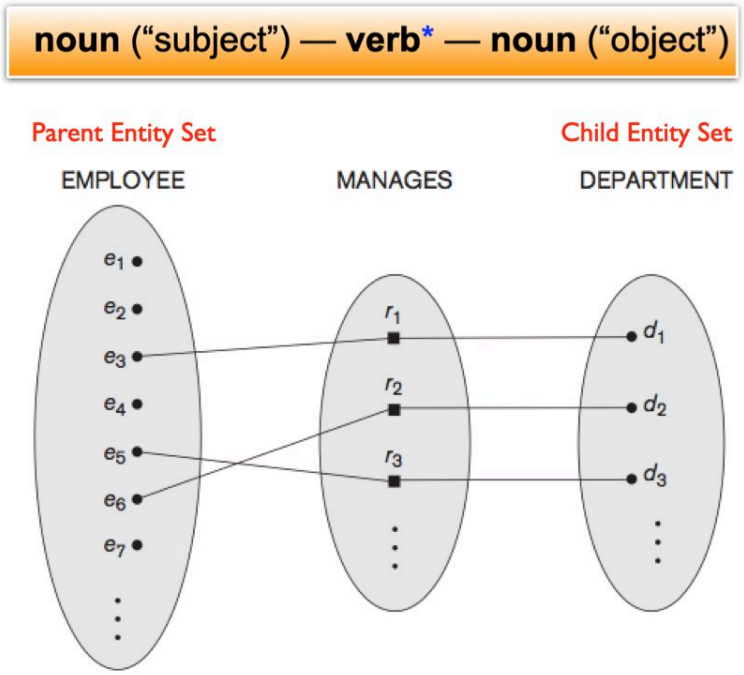
  Example:

  1. STUDENTS – (MUST) REGISTER TO – CLASS
  2. STUDENTS – (CAN) HAVE MAXIMUM TWO – ADVISORS
  3. PERSON – (MAY) OWN A CAR / (MAY) BELONG TO – POLITICALPARTY
  4. ...



Example





* with its “modality” (modal verbs)





Remember: The starting point...

- Conceptual Design: The designer refers to data models, and by applying the concepts of the chosen model, translates the specified requirements into a conceptual schema (view) of the database.
- So, there is a direct correspondence between requirements and (conceptual) design.
- In particular, conceptual design is easier if the previous phase of **requirements analysis** is carried by a data model in mind.



(Chen, 1983)





Data Models - The Relational Model

- A data model is a collection of tools for describing
  - Data and Data relationships
  - Data semantics
  - Data constraints
- Relational model (*attributes* and *tables* = *sets of rows*)





Data Models - The E-R Model

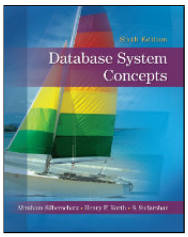
- A data model is a collection of tools for describing
  - Data and Data relationships
  - Data semantics
  - Data constraints
- Relational model (*attributes* and *tables* = *sets of rows*)
- E-R model (*attributes of entities* and *relationships*)



The E-R Model

i.e.: The Entity-Relationship Data Model

cf. Chapter 7 T, sections 7.1–7.5



Premise: Codd's Model (1970)

Suggested reading (non-technical parts):

```
@Article{Codd70cacm,
  author      = "E. F. Codd",
  title       = "A Relational Model of Data for Large Shared Data Banks",
  journal     = "Communications of the ACM",
  year        = "1970",
  volume      = "13",
  number      = "6",
  pages       = "377--387"
}
```





Chen's Model (1976)



Suggested reading (non-technical parts):

```
@Article{Chen76tods,
  author      = "Peter Pin-Shan Chen",
  title       = "The Entity-Relationship Model—Toward a Unified View of Data",
  journal     = "ACM Transactions on Database Systems (TODS)"
  year        = "1976",
  month       = "March",
  volume      = "1",
  number      = "1",
  pages       = "9--36"
}
```



The Entity-Relationship Model—Toward a Unified View of Data

PETER PIN-SHAN CHEN

Massachusetts Institute of Technology

A data model, called the entity-relationship model, is proposed. This model incorporates some of the important semantic information about the real world. A special diagrammatic technique is introduced as a tool for database design. An example of database design and description using the model and the diagrammatic technique is given. Some implications for data integrity, information retrieval, and data manipulation are discussed.

The entity-relationship model can be used as a basis for unification of different views of data: the network model, the relational model, and the entity set model. Semantic ambiguities in these models are analyzed. Possible ways to derive their views of data from the entity-relationship model are presented.

Key Words and Phrases: database design, logical view of data, semantics of data, data models, entity-relationship model, relational model, Data Base Task Group, network model, entity set model, data definition and manipulation, data integrity and consistency
CR Categories: 3.50, 3.70, 4.33, 4.34

1. INTRODUCTION



Chen's Model (*cont.*)

"The logical view of data has been an important issue in recent years. Three major data models have been proposed: the network model …, the relational model …, and the entity set model…

These models have their own strengths and weaknesses. …

This paper presents the entity-relationship model, which has most of the advantages of the above three models. The entity-relationship model adopts the more natural view that the real world consists of entities and relationships." (Chen, 1976)





From Tables to Entity Sets

- **Example:** Tables ("relations") in relational model:
- Note 1: attribute *dept_name* is repeated
- Note 2: *dept_name* in *instructor* represents a relationship…

instructor:

ID	*name*	*dept_name*	*salary*
22222	Einstein	Physics	95000
12121	Wu	Finance	90000
32343	El Said	History	60000
45565	Katz	Comp. Sci.	75000
98345	Kim	Elec. Eng.	80000
		y	72000
		. Sci.	65000
		y	62000
		. Sci.	92000
			40000
		s	87000
		e	80000

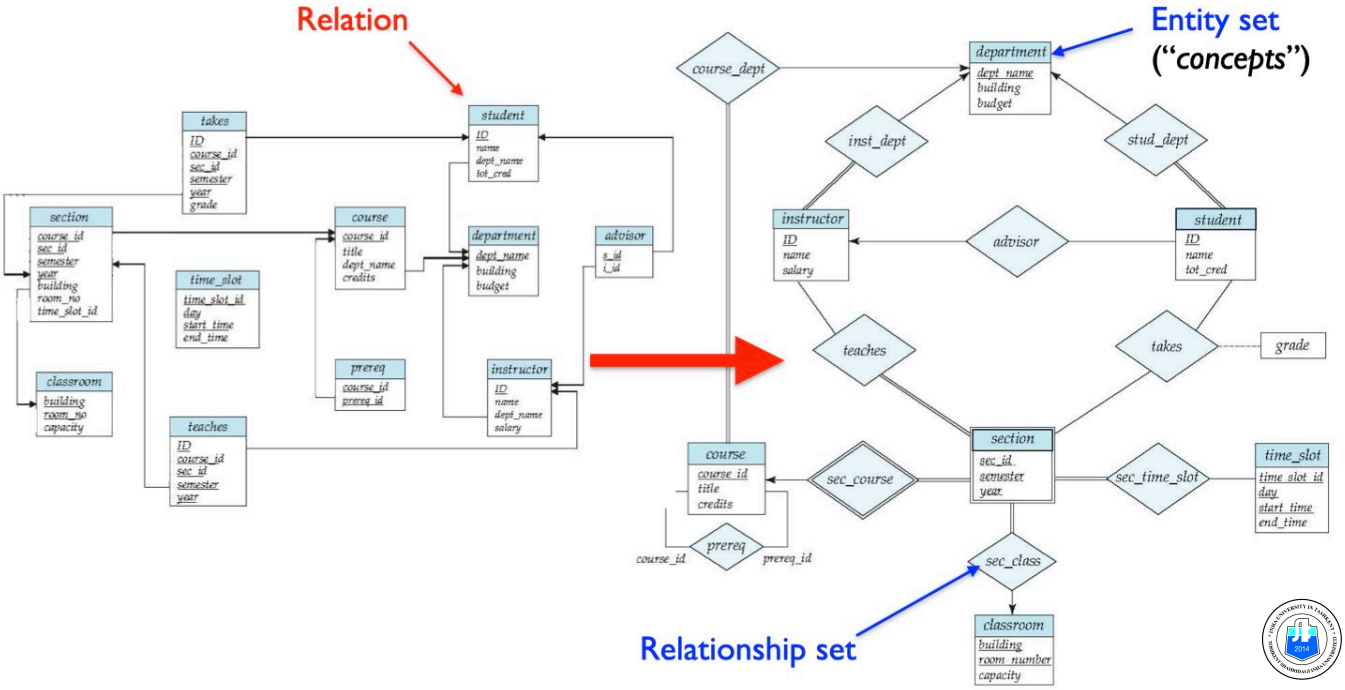
department:

dept_name	*building*	*budget*
Biology	Watson	90000
Comp. Sci.	Taylor	100000
Elec. Eng.	Taylor	85000
Finance	Painter	120000
History	Painter	50000
Music	Packard	80000
Physics	Watson	70000





Relation *Schemas* versus E-R *Diagrams*



Outline

1. From Previous Week
2. Conceptual Data Design
3. Data and Data Relationships
4. Data Semantics
5. Data Constraints
6. Entity Integrity *Revised*: Weak Entities
7. Example: Company DB
8. LAB #1 : E-R Diagramming

Data and Data Relationships

or: How we structure data in the E-R Model

- **Data and Data relationships**
- **Data semantics**
- **Data constraints** (for business rules)



E-R model -- Database Modeling

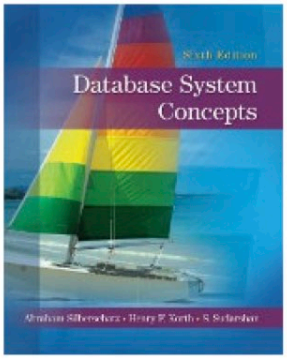
- The E-R model employs three basic concepts:
  - entity set,
  - relationship set,
  - attribute (of entity and/or relationship sets).
- The E-R model has an associated diagrammatic representation, the E-R diagrams, which allows to express the logical structure of a DB graphically.

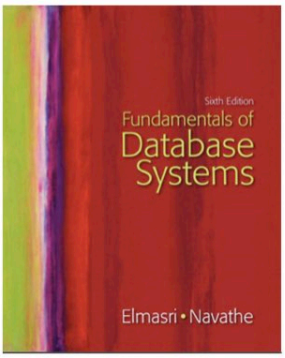




Remark on Terminology

- Terminology in Textbook (T) and Reference book (R):





- An **entity set** in T is **entity type** in R.
- A **relationship set** in T is **relationship type** in R.



Entity Sets

or, equivalently: Entity Types



Entities and Entity Sets (Types)

- An **entity** is a "thing" or "object" that exists and is distinguishable from other things and objects.
  - Example: a person, company, event, plant
- An **entity set (or type)** is a set of entities of same type that share the same properties (or: attributes).
  - Example: set of persons, companies, trees, holidays
- The **extension** of an entity set is the actual collection of entities belonging to the entity set.
  - Example: actual instructors form the extension of the entity set *instructor*.





Entity Sets (Types) - Example

- The extension of entity sets *instructor* and *student*:

76766	Crick
45565	Katz
10101	Srinivasan
98345	Kim
76543	Singh
22222	Einstein

instructor

98988	Tanaka
12345	Shankar
00128	Zhang
76543	Brown
76653	Aoi
23121	Chavez
44553	Peltier

student





Entity Sets (Types) - Attributes

98988	Tanaka
12345	Shankar
00128	Zhang
76543	Brown
76653	Aoi
23121	Chavez
44553	Peltier

student

- An entity set (type) is defined by a unique name and a set of **attributes**; i.e., descriptive properties possessed by all members of the set.
  - Example: Textual notation for entity sets *instructor* and *course*:

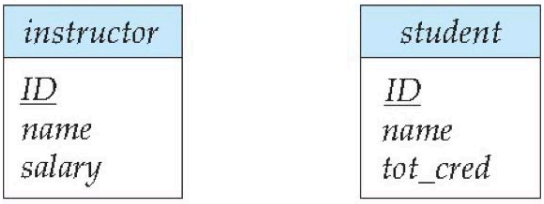
    instructor (*ID, name, street, city, salary*)
    course (*course_id, title, credits*)
- ***NB**: what* attributes to capture in the database is basically a business-based design decision.
- Attributes have a **value** (instance of the attribute)
- A subset of attributes forms a **primary key** of the entity set and uniquely identifies an entity in the set.





Entity Sets: Diagramming

- Entities can be represented graphically as follows:
  - Rectangles represent entity sets.
  - Attributes listed inside entity rectangle
  - Underline indicates primary key attributes







More on Attributes

- An attribute's **domain** is the set of permitted values for the attribute
- Attribute types:
  - **Simple** and **composite** attributes.
  - **Single-valued** and **multivalued** attributes
    - Example: multivalued attribute: *phone_number*
  - **Derived** attributes
    - Can be computed from other attributes
    - Example: *age*, given *date_of_birth*
  - Attributes may have value *null*

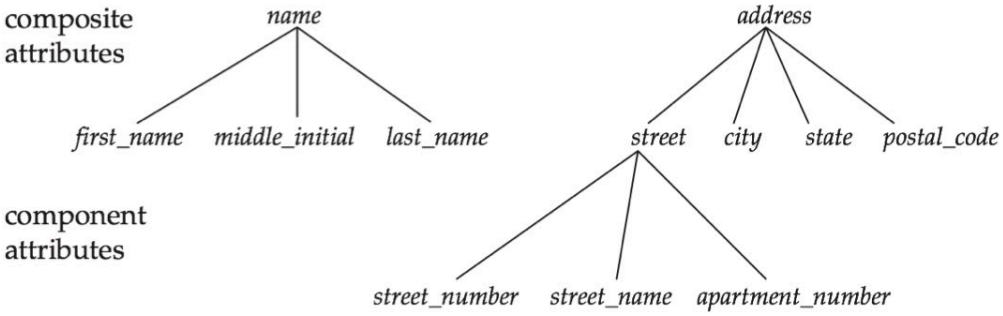
Notation : *{phone_number}*





Composite Attributes

- An attribute's **domain** is the set of permitted values for the attribute
- Attribute types:
  - **Simple** and **composite** attributes.

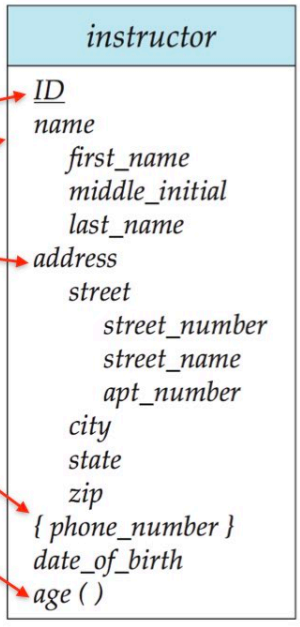






Attributes of Entity Sets: Diagramming

- Attributes representation:
  - **Simple** and **key**
  - **composite**
  - **multivalued**
  - **Derived**





Relationship Sets

or, equivalently: Relationship Types



Relationship Sets (Types)

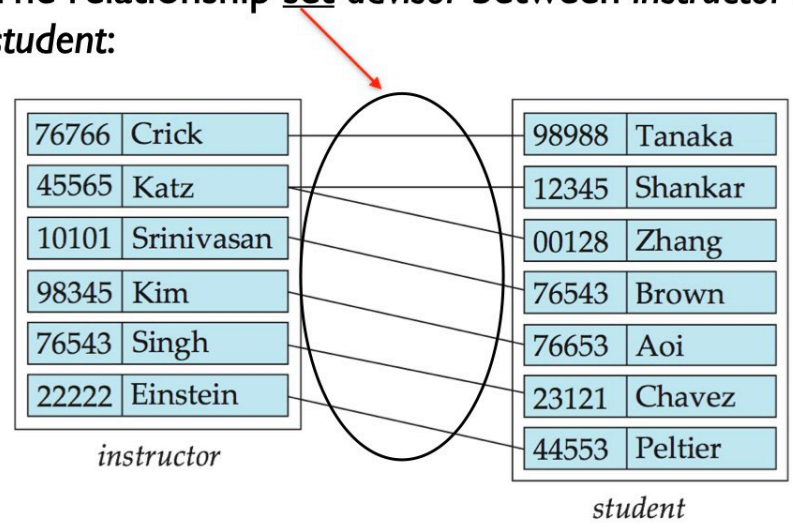
- A **relationship** is an association among entities
  - Example:
    advisor((22222,Einstein),(44553,Peltier))
- A **relationship set (or type)** is a set of a mathematical relations among $n \geq 2$ entities, each taken from some entity sets E_1 E_2, …, E_n, i.e.:
  $R = \{(e_1, e_2, \ldots e_n) \mid e_1 \in E_1, e_2 \in E_2, \ldots, e_n \in E_n\}$
  - E.g.: ((44553,Peltier),(22222,Einstein)) $\in$ *advisor*
- The **degree of a relationship set** R is the number of entity sets E_i that "participate" in R.





Relationship Sets - Example

- The relationship set *advisor* between *instructor* and *student*:

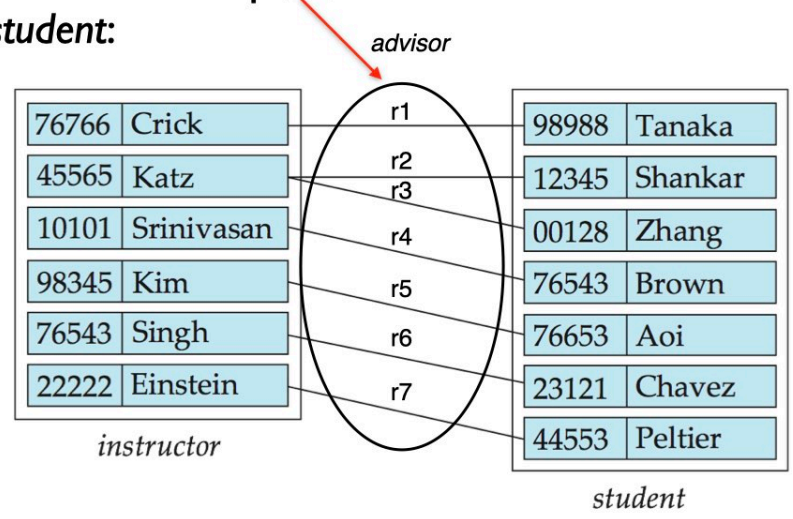






Relationship Sets - Example (*cont.*)

- The relationship set *advisor* between *instructor* and *student*:



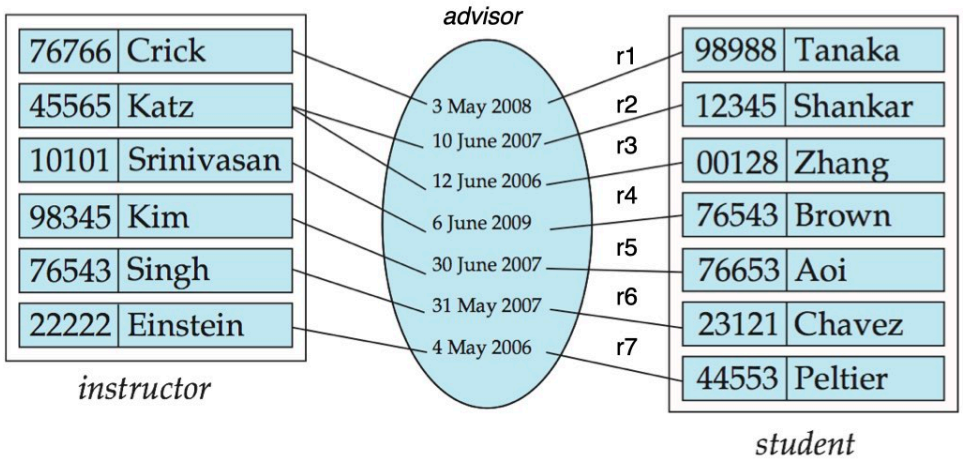
advisor = {r1, r2, r3, r4, r5, r6, r7}





Relationship Sets - Attributes

- Also relationship sets can have some **attributes**.
  - Example: relationship set *advisor* with attribute *date*:

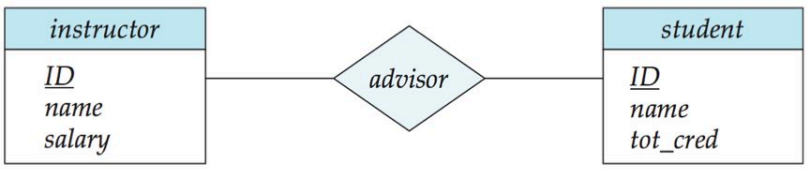






Relationship Sets: Diagramming

- Diamonds represent relationship sets.

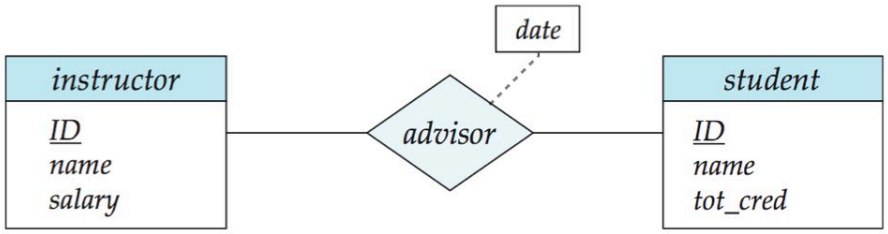






Relationship Sets: Diagramming

- Relationship sets with attributes ("date") are represented this way:

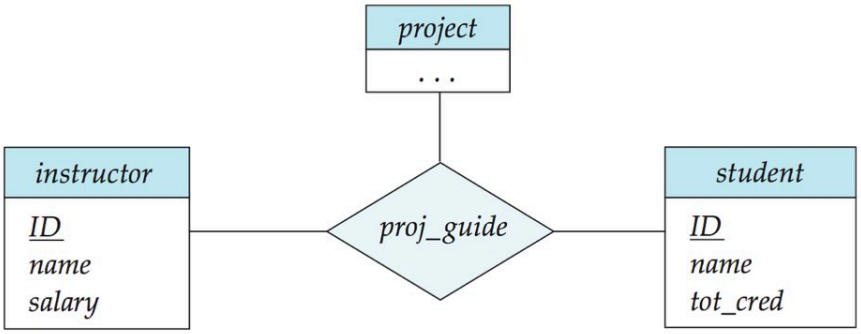






Relationship Sets: Diagramming

- Ternary relationship sets are represented this way:

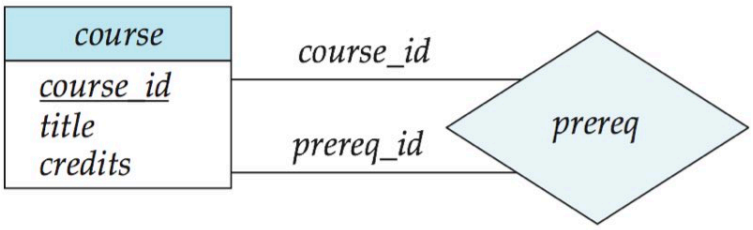






Unary Relationship Sets — Roles

- Entity sets of a relationship need not be distinct
  - Each occurrence of an entity set plays a "role" in the relationship set.
- The labels "*course_id*" and "*prereq_id*" are called **roles**.





Outline

1 From Previous Week
2 Conceptual Data Design
3 Data and Data Relationships
4 Data Semantics
5 Data Constraints
6 Entity Integrity *Revised*: Weak Entities
7 Example: Company DB
8 LAB #1 : E-R Diagramming

Data Semantics

or: How we mean data in the E-R Model

- **Data and Data relationships**
- **Data semantics**
- **Data constraints** (for business rules)



More on Attributes (*cont.*)

- An attribute's **domain** is the set of permitted values (of a certain "type") for the attribute.
- Type of attribute defines data semantics (meaning).
- Data semantics is a "grounded" into data types.
- Data types depend on DBMS and language used to implement the design.

- Attributes values may have value *null* :
  - the value "unknown", or
  - the value does not exist.



Outline

1. From Previous Week
2. Conceptual Data Design
3. Data and Data Relationships
4. Data Semantics
5. Data Constraints
6. Entity Integrity *Revised*: Weak Entities
7. Example: Company DB
8. LAB #1 : E-R Diagramming

Data Constraints

or: How we capture constraints on data in the E-R Model (and why)

- **Data and Data relationships**
- **Data semantics**
- **Data constraints** (technical and business rules)

Entity Integrity

or: Keys for Entity Sets



Keys for Entity Sets = Keys for Schemas

- **Def**: A **super-key** for an entity set *E* is a set of one or more attributes of E that, taken collectively, allow us to identify uniquely an entity in *E*.

- **Example**: Suppose *E* = *student* and
  student = *student* (*ID*,*name*,*dept_name*,*tot_cred*).

  Suppose *ID* attribute is sufficient (by design!!!) to distinguish one *student* from another. Then,
  - *ID* is a super-key (in short: SK) for *student*.
  - **NB**: you can decide that {*dept_name*, *tot_cred*} is SK.





Keys for Entity Sets = Keys for Schemas

- **Remark**: If K is a super-key for an entity set, then any superset of K is a super-key.
  - **Example**: In
    student (*ID*,*name*,*dept_name*,*tot_cred*),
    attribute *ID* super-key implies that also, say,
    {*ID*,*name*} and {*ID*,*dep_name*}
    are a super-key for *student.*
  - Best super-keys are minimal, i.e., do not have a proper subset that is a super-key.
  - Minimal super-keys are called **candidate keys**.





Keys for Entity Sets = Keys for Schemas

- **Def**: A **primary key** for an entity set *E* is a candidate key chosen by the database designer to uniquely identify/distinguish all entities in *E*.
- **Remark 1**:
  a primary key is for an entity set.
- **Remark 2**:
  Primary keys must be chosen with care. (*)

  (*) *we will see more on it soon*



Relation-Based Constraints (I)

Mapping Cardinality (or: Cardinality Ratio)



Mapping Cardinalities

- Express the number of entities to which another entity can be associated via a relationship set.
- Most useful in describing binary relationship sets.
- For a binary relationship set the mapping cardinality must be one of the following types:
  - One-to-one, or: cardinality ratio 1:1
  - One-to-many, or: cardinality ratio 1:N (*)
  - Many-to-one, or: cardinality ratio N:1
  - Many-to-many, or: cardinality ratio M:N

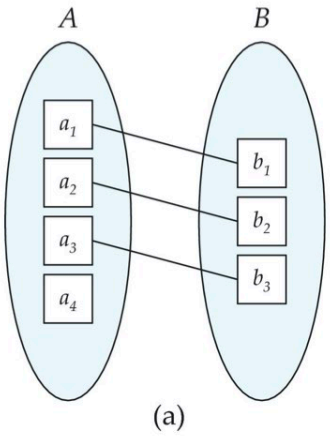
(*) N means *many* or *any number* (zero or more)





Mapping One-to-One (Ratio 1:1)

- An entity in *A* is associated with *at most one* entity in *B*, and vice versa.



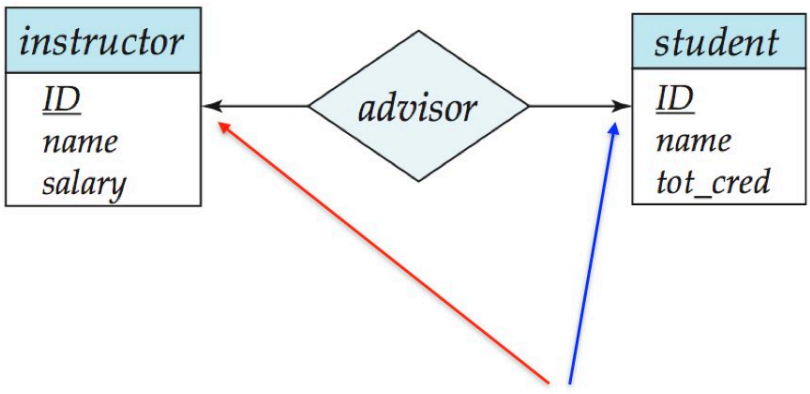
One-to-one





1-1 Relationships: Diagramming

- Relationship sets with mapping cardinality one-to-one ("1-1" in short) are represented this way:

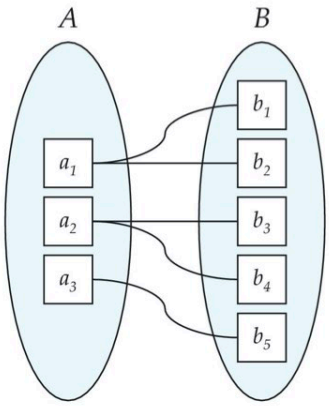






Mapping One-to-Many (Ratio 1:N)

- An entity in *A* is associated with *any number* (0..*) of entities in *B*; an entity in *B* is associated as before.



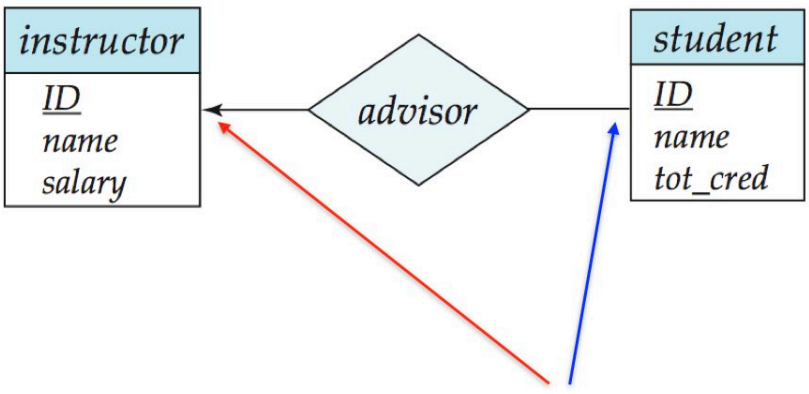
One-to-many





1-N Relationships: Diagramming

- Relationship sets with mapping cardinality one-to-many ("1-N") are represented this way:

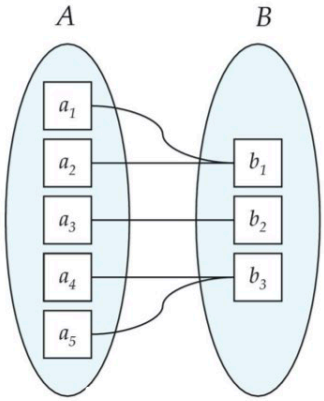






Mapping Many-to-One (Ratio N:1)

- An entity in *A* is associated with *at most one* entity in *B*; an entity in *B* is associated with 0..* entities in *B*.



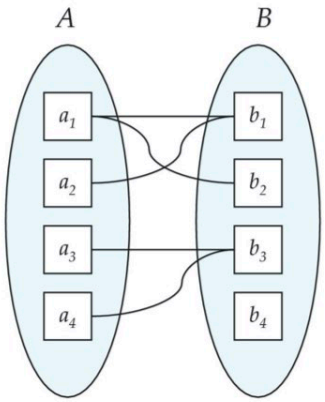
Many-to-one





Mapping Many-to-Many (Ratio M:N)

- An entity in *A* is associated with *any number* (0..*) of entities in *B*, and vice versa.



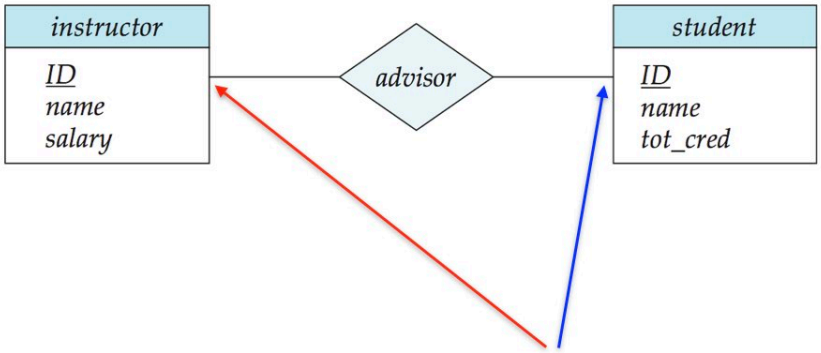
Many-to-many



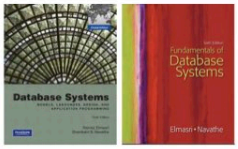


M-N Relationships: Diagramming

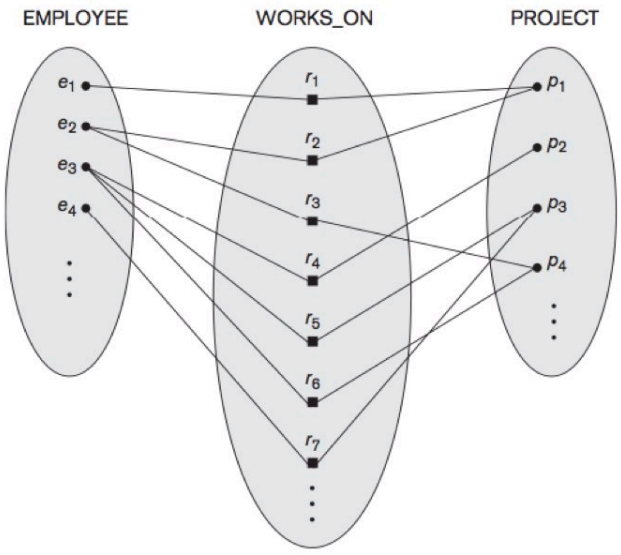
- Relationship sets with mapping cardinality many-to-many ("M-N") are represented this way:







Example





Relation-Based Constraints (II)

Entity Participation



Total and Partial Participation

- In the relationship set

$$R = \{(e_1, e_2, \ldots e_n) \mid e_1 \in E_1, e_2 \in E_2, \ldots, e_n \in E_n\}$$

the entity sets $E_1, E_2, \ldots, E_n$ participate in R.
Then, we also say that $e_1, e_2, \ldots e_n$ participate in R.

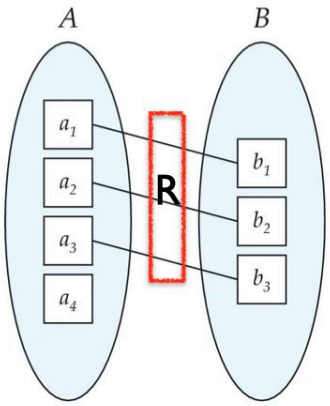
- The participation of E_i in R is **total** if every entity $e_{ij} \in E_i$ participates in *at least one* relationship in R.
- Otherwise, the participation of E_i in R is **partial**.





Example

- The participation of entity set *B* in the relationship set R = $\{(a_1, b_1),(a_2, b_2),(a_3, b_3)\}$ is total.

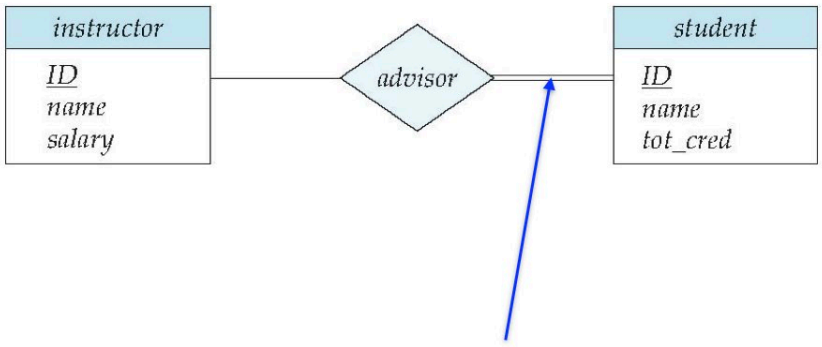






Total Participation: Diagramming

- Relationship sets with total participation of an entity set are represented this way (see "*student*" side):

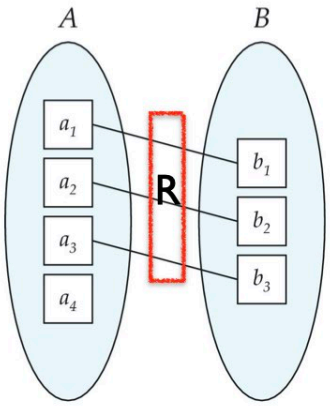






Example (*cont.*)

- The participation of entity set *A* in the relationship set R = $\{(a_1, b_1),(a_2, b_2),(a_3, b_3)\}$ is partial.

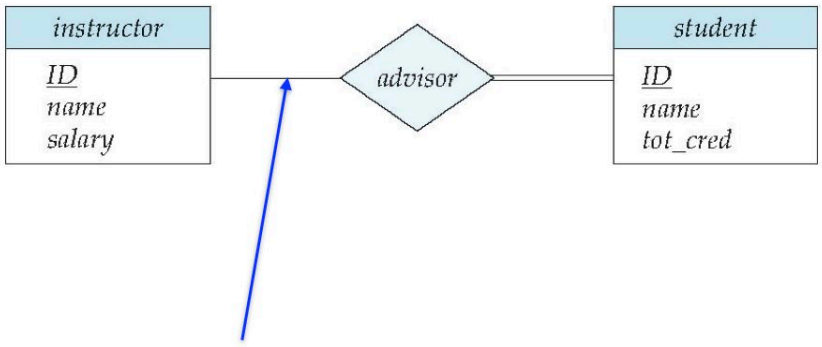






Partial Participation: Diagramming

- Relationship sets with total participation of an entity set are represented this way (see "*instructor*" side):

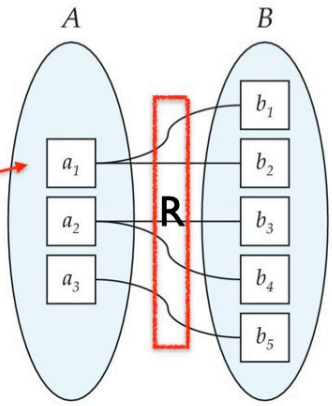






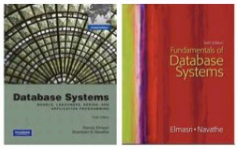
Example (*cont.*)

- The participation of both entity sets *A* and *B* in the relationship set R = $\{(a_1, b_1),(a_1, b_2),(a_2, b_3),(a_2, b_4),(a_3, b_5)\}$ is total.



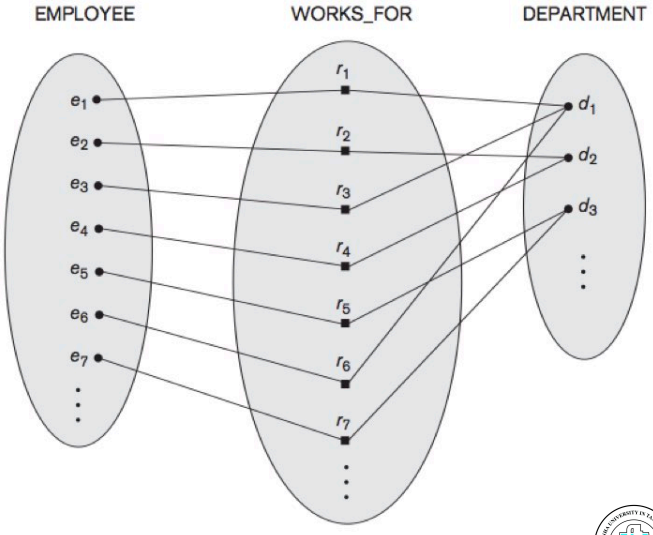
Note : entity a_1 participates in two relationships in R, but to define the participation of A in R to be **total** it is enough that a_1 (and **all** the other entities in A) participates in at least one relationship in R.





Example

- If *every* employee must work for a department, then an employee entity can exist only if it participates in at least one *works_for*'s relationship.
- participation of *employee* in *works_for* is total.





Structural constraints

- In summary:
  - mapping cardinality (or: cardinality ratio) of entity sets participating in a relationship set R - 1:1, 1:N, N:1,M:N
    - specifies *maximum* participation;
  - participation constraint (or: existence dependency) of an entity set in a relationship set R - partial / total
    - specifies *minumum* participation

zero (optional participation, not existence-dependent)

one or more (mandatory participation, existence-dependent)

Together are called structural constraints of R.





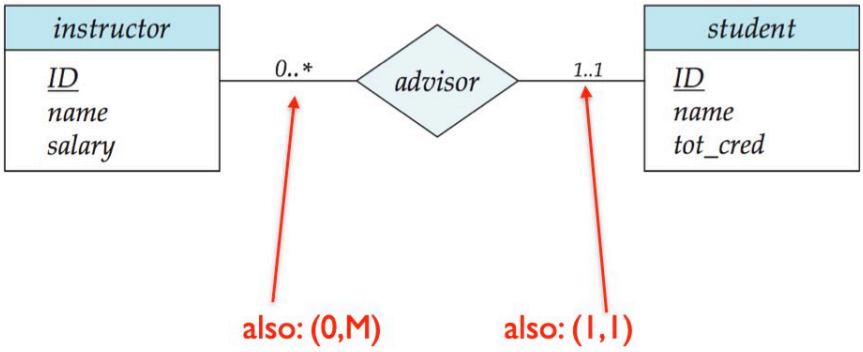
(Min,Max) Participation: *Look-here* approach

- A **single line** may have an associated minimum and maximum cardinality, shown in the form *l..h* or (*l*,*h*), where *l* is minimum and *h* maximum cardinality.
- A minimum of 0 indicates partial participation.
- A minimum of 1 indicates total participation.
- A maximum of 1 indicates that the entity participates in at most one relationship.
- A maximum of * (or a letter) indicates that the entity participates in the relationship with no limits.





(Min,Max) Participation (*cont.*): Notation

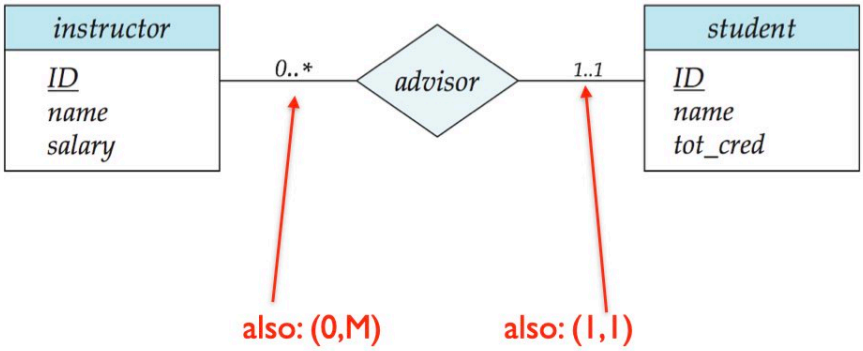






(Min,Max) Participation: Example

- *Instructor* can advise 0 or more students.
- A student must have 1 advisor; cannot have multiple advisors.







Participation's Roles

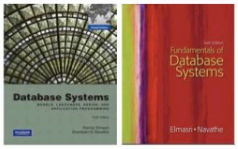
- In the relationship set R defied as

$$R = \{(e_1, e_2, \ldots e_n) \mid e_1 \in E_1, e_2 \in E_2, \ldots, e_n \in E_n\},$$

the entity sets $E_1, E_2, \ldots, E_n$ participate in R.

- The function that an entity has or "plays" in a relationship, e.g. $e_1 \in E_1$ in relationship $(e_1, e_2, \ldots e_n)$ within the relationship set R above is called **role.**
  - Example:
    see the next slide

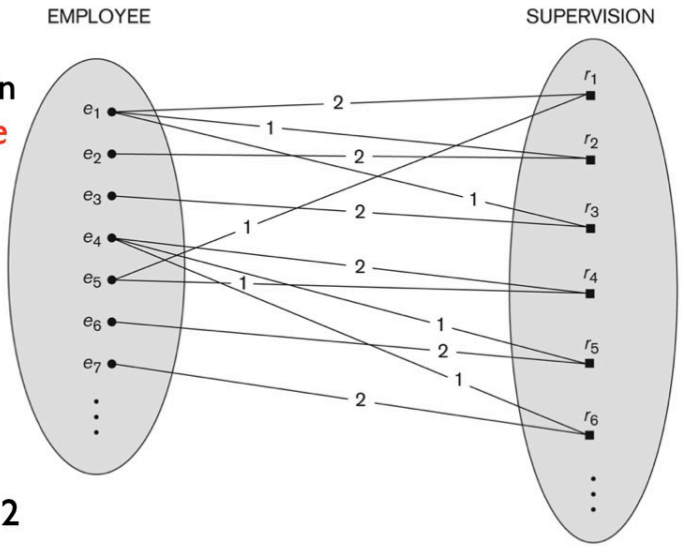




Example (on Roles)

- An EMPLOYEE participates in SUPERVISION (i.e., a *recursive relationship set*) twice in two distinct roles:
  1. supervisor (boss)
  2. supervisee (subordinate)

  (called *recursive participation*)
- In figure: boss's participation is labeled 1, subordinate's participation is 2





Relationship Integrity

or: Keys for Relationship Sets



Keys for Relationship Sets

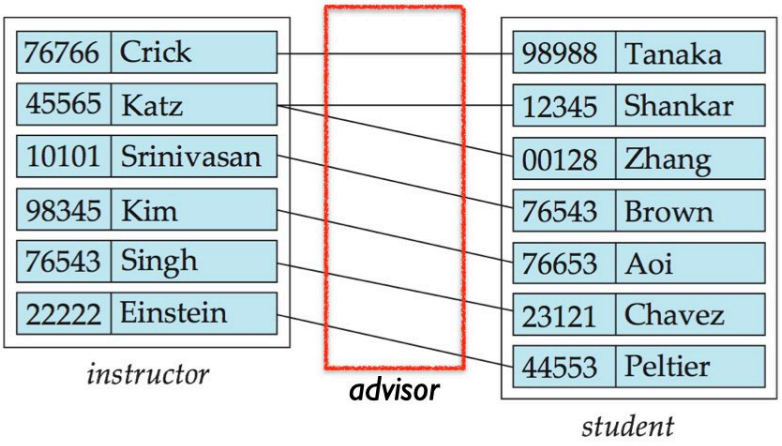
- Let relationship set
  $R = \{(e_1, e_2, \ldots e_n) \mid e_1 \in E_1, e_2 \in E_2, \ldots, e_n \in E_n\}$
  be given. Let *primary-key*(E_i) denote the set of attributes that forms the primary key for E_i.

- To define primary key for R, we have two cases:
  - 1: the attribute names of primary keys are unique across all entity sets E_i in R.
  - 2: the attribute names of primary keys are not unique across all entity sets E_i in R.





Example



- *primary-key(instructor)={ID}*
  primary-key(student)={ID}





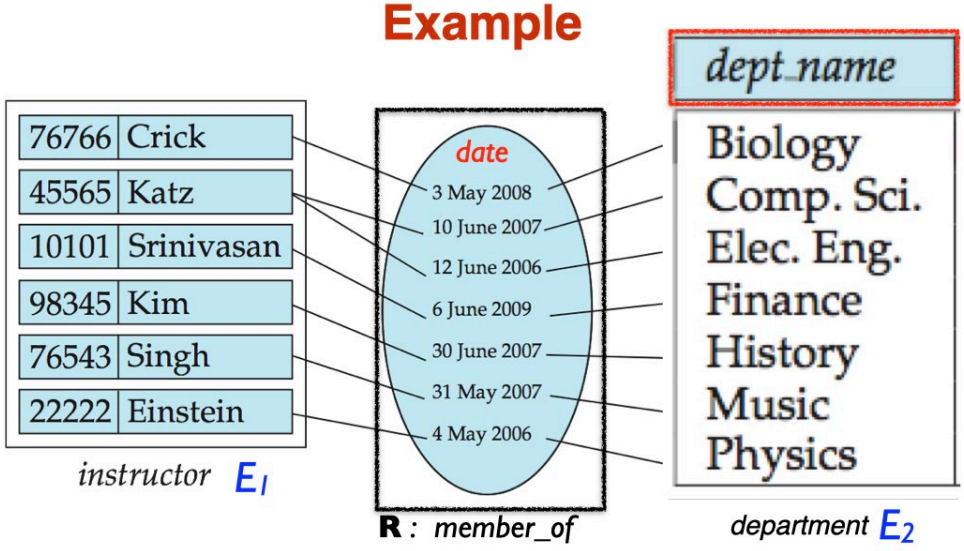
Keys for Relationship Sets (*cont.*)

- **Case 1**: attribute names of primary keys are unique.
  - If R has no descriptive attributes, then the set
    $primary\text{-}key(E_1) \cup \ldots \cup primary\text{-}key(E_n)$ (*)
    describes uniquely any relationship in R.
  - Otherwise, let $a_1, \ldots a_m$ be R's attributes. Then:
    $primary\text{-}key(E_1) \cup \ldots \cup primary\text{-}key(E_n) \cup \{a_1, \ldots a_m\}$
    describes uniquely any relationship in R.
  - In both cases, (*) is a super-key for R.
  - Primary key for R depends of mapping cardinality!





Example



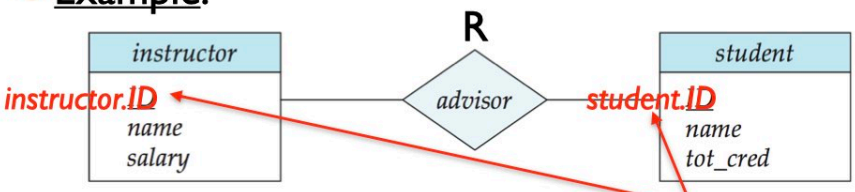
- $primary\text{-}key(E_1) \cup primary\text{-}key(E_2) \cup \{a_1, \ldots a_m\} =$
  $= \{ID\} \cup \{dept_name\} \cup \{date\}$ is a super-key for R.





Keys for Relationship Sets (*cont.*)

- **Case 2**: attribute names of primary keys are <u>not unique</u> across all entity sets E_i in R.
  - <u>Example</u>:

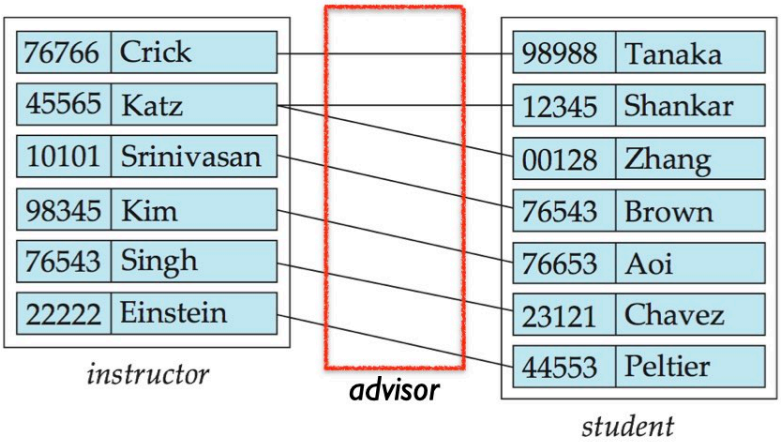


- Not unique names are renamed;
  - the name of the entity set combined with the name of the attribute forms a <u>unique name</u>.
- Then Case 1 applies.





Example



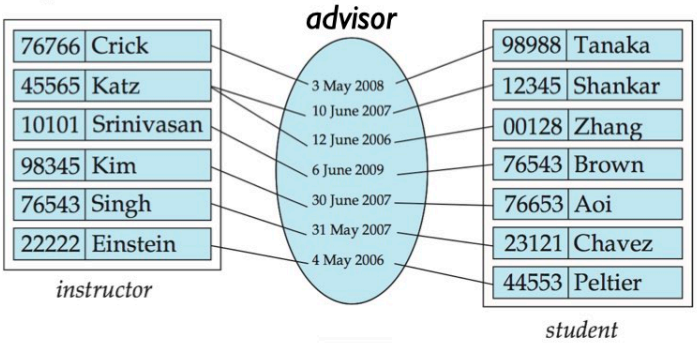
- *primary-key(instructor)={instructor.ID}*
  primary-key(student)={student.ID}



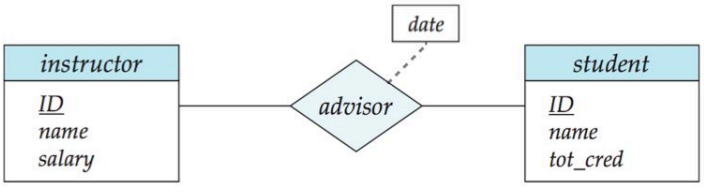


Primary Keys for Relationship Sets

- **Example**: Consider *advisor* with attribute *date*:



E-R diagram:
(*we'll see more later*)







Primary Keys for Relationship Sets (*cont.*)

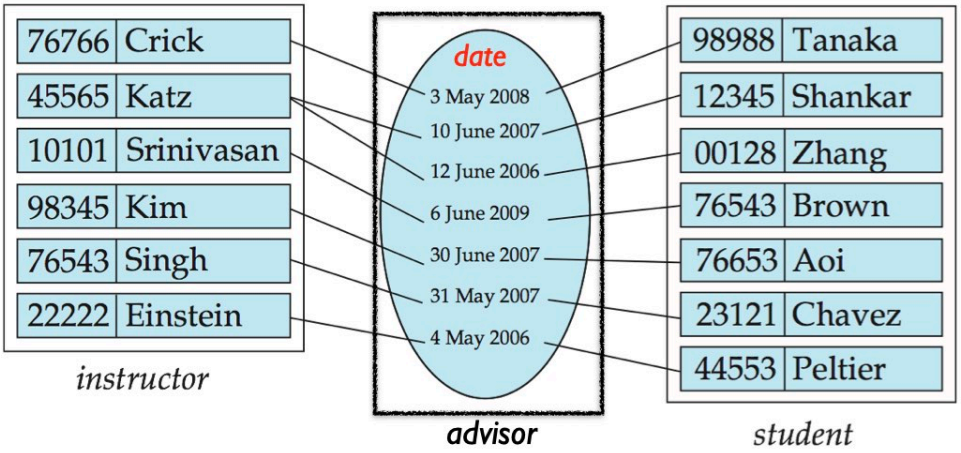
- A super-key for *advisor* is:
  primary-key(instructor) ∪ *primary-key(student)* =
  = {*instructor.ID*, *student.ID*}.
  - Suppose *advisor* is many-to-many. Then:
    primary key for *advisor* = {*instructor.ID*, *student.ID*}.
  - Suppose *advisor* is many-to-one from *student* to *instructor*: primary key for *advisor* = {*student.ID*}.
  - ✓ Suppose *advisor* is many-to-one from *instructor* to *student*: primary key for *advisor* = {*instructor.ID*}.
  - ✓ Suppose *advisor* is one-to-one. Then:
    primary key for *advisor* is *instructor.ID* or *student.ID*.

designer's decision





Example (cont.)



- Mapping cardinality is many-to-one from *student* to *instructor*: primary key for *advisor* is {*student.ID*}.



Outline

1. From Previous Week
2. Conceptual Data Design
3. Data and Data Relationships
4. Data Semantics
5. Data Constraints
6. Entity Integrity *Revised*: Weak Entities
7. Example: Company DB
8. LAB #1 : E-R Diagramming

Entity Integrity *Revised*

or: Keys for *Weak* Entity Sets



Weak Entity Sets

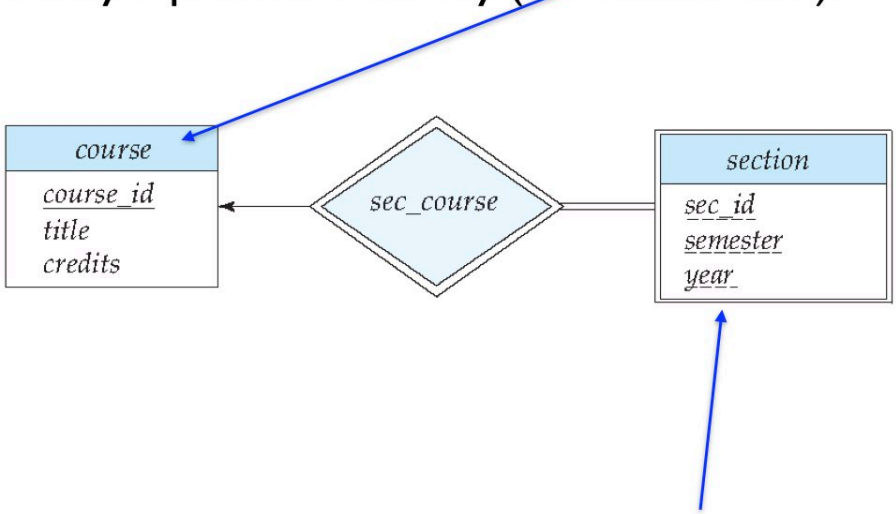
- An entity set that does not have sufficient attributes to form a primary key is termed a **weak entity set**.
  - Otherwise, it is termed a (**strong**) **entity set**.
- For a weak entity set to be meaningful, it must be associated with another entity set, called the **identifying** or **owner entity set**.
- A weak entity is associated with an identifying entity; so it is said to be **existence dependent** on the identifying entity set.
- Relationship associating the weak entity set with the identifying entity set is **identifying relationship**.





Weak Entity Sets: Diagramming

- Weak entity sets together with their **owners** are usually represented this way (see "*section*" side):



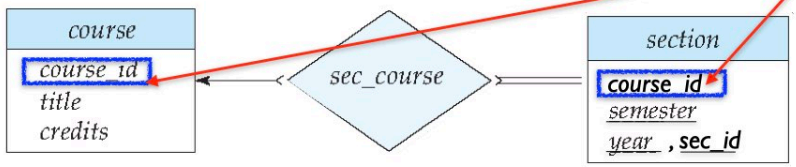




Example

- Consider a *section* entity, which is uniquely identified by a course identifier, semester, year, and section identifier.
  - Section entities are related to course entities.
  - Suppose we create a relationship set *sec_course* between entity sets *section* and *course*.
  - Observe: information in *sec_course* is redundant.

E-R diagram: (*we'll see more later*)

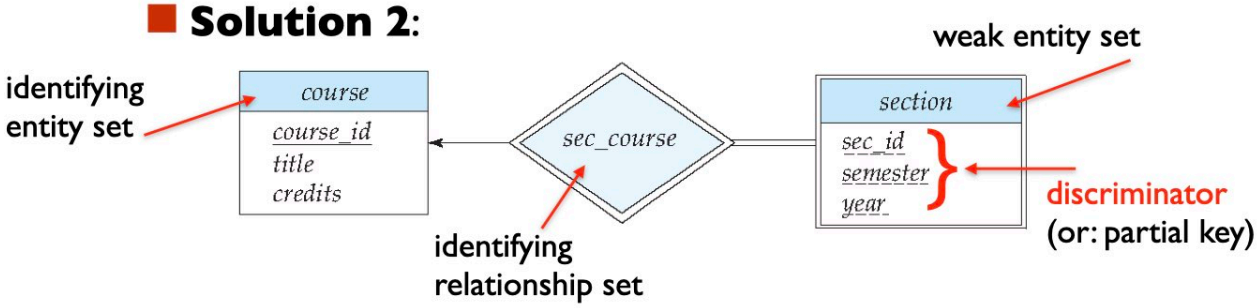






Example (*cont.*)

- **Solution 1**: not to use relationship set *sec_course*
  - not desirable (relationship between *section* and *course* becomes implicit in an attribute - *course_id*)
- **Solution 2**:



- Primary key of *section* is: {*course_id*} ∪ *discriminator*.



Outline

1. From Previous Week
2. Conceptual Data Design
3. Data and Data Relationships
4. Data Semantics
5. Data Constraints
6. Entity Integrity *Revised*: Weak Entities
7. Example: Company DB
8. LAB #1 : E-R Diagramming

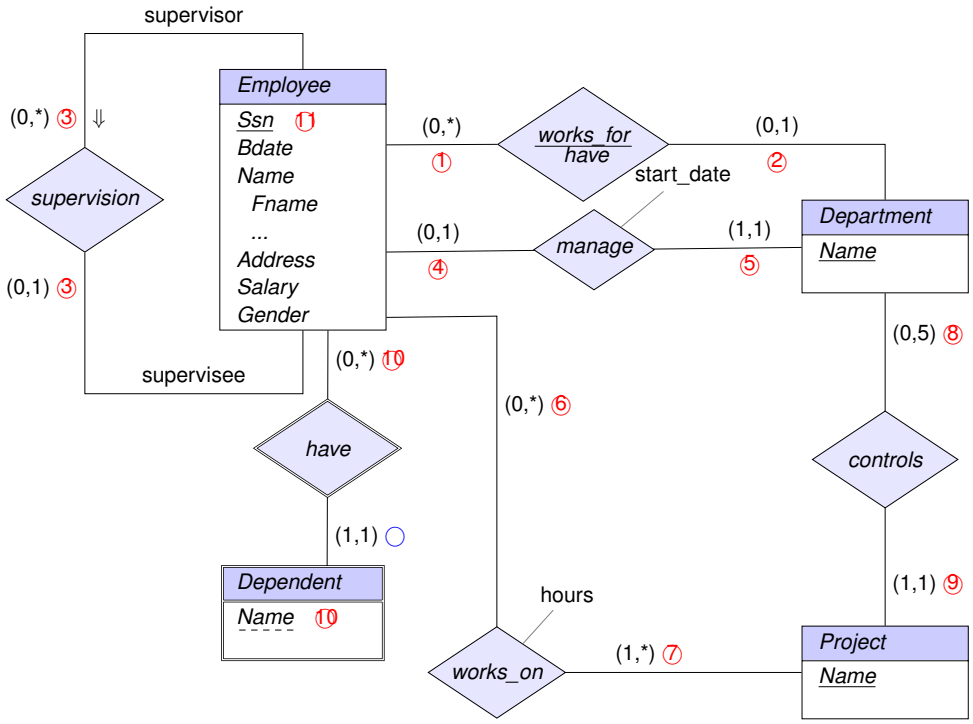
Example

Company DB Conceptual Design

CompanyDB: Draw an equivalent E-R diagram (with Primary Keys!)

1. Each employee must work for at most one department.
2. A department must have at least one employee (working for it).
3. Employees can supervise other employees and be supervised by at most one employee.
4. An employee can manage at most one department, and start date is stored.
5. Every department must be managed, and by exactly one employee. The start date of managing is stored.
6. Every employee can work on a project a number of hours.
7. A project must have at least one employee working on it (for some hours).
8. A department can control many projects, but no more than five.
9. All projects are controlled by one and only department.
10. Employees can have dependents, weak entities identified by name.
11. Employees have a unique "social security number" (*ssn*), day of birth (*Bdate*), a composite name (with components: *Fname* for first name,...), address, salary, and gender.

A solution (numbers refer to each sentence)



Outline

1. From Previous Week
2. Conceptual Data Design
3. Data and Data Relationships
4. Data Semantics
5. Data Constraints
6. Entity Integrity *Revised*: Weak Entities
7. Example: Company DB
8. LAB #1 : E-R Diagramming

LAB #1: *Conceptual* Data Design*

* Participation checked (2-3 members per working group)

In class only (no slides provided).



Summary

1 From Previous Week

2 Conceptual Data Design

3 Data and Data Relationships

4 Data Semantics

5 Data Constraints

6 Entity Integrity *Revised*: Weak Entities

7 Example: Company DB

8 LAB #1 : E-R Diagramming



DATABASE

End of Slides

WEEK 3

Code: W3.L1L2[LAB#1]